

Week 3 Introduction

Mathematics for Multidimensional Data

Working with Multidimensional Data in Python



Video: 3.7 Basic Plotting

2 min



Reading: Note on Matplotlib

10 min



Practice Quiz: Basic

Plotting – Review

Information

3 questions



Reading: Matplotlib Scatter

Plot Documentation

20 min



Video: 3.7a Storing 2D

Coordinates in a Single Data
Structure

6 min



Practice Quiz: Storing 2D

Coordinates – Review

Information

2 questions



Video: 3.8 Multidimensional

Mean

4 min



Practice Quiz:

Multidimensional Mean –

Review Information

2 questions



Video: 3.9 Adding Graphical

Overlays

5 min



Practice Quiz: Adding

Graphical Overlays – Review

Information

3 questions



Reading: Matplotlib Patches

Documentation

10 min



Video: 3.10 Calculating the

Distance to the Mean



3.12 Errata

At around 4'10 in the previous video, you can see Matthew being fully in 'min' and 'max'. This is in fact considered bad practice, and he would certainly not recommend that happens to you, it is something any coder, even seasoned ones, know to avoid (double check your code and proofreading yourself!). The problem happening here is that the functions `min()` and `max()`, that we have used in the course. Once you update your code, they will no longer be accessible!

Try that in your code, it is a good experiment (you can run all three examples below):

1. `min()` can be used as a function:

```
1 x = [1,2,3]
2 print(min(x))
```

1

2. Min is redefined as a variable, it can be printed out:

```
1 x = [1,2,3]
2 min = 5
3 print(min) # min now is a variable holding the value 5
```

5

3. Min is now a variable, it cannot be used as a function any more!

```
1 x = [1,2,3]
2 min = 5
3 print(min(x)) # since min is a variable with value 5, the min()
4
```

Error on line 3:

```
print(min(x)) # since min is a variable with value 5, the min()
TypeError: 'int' object is not callable
```